

Q1. 8085 Program Execution

Aim: Examine the execution of 8085 instruction

Observation

BEFORE Execution

PC, SP and memory values change during execution

NEXT SCREENSHOT

Arithmetic and branch instruction execute correctly

Conclusion

The program executes correctly and shows the instruction execution sequence.

Q2. Bitwise operations

Aim perform AND, OR, XOR, and NOT operations

observation

The hexadecimal result match the expected bitwise operations.

conclusion

Bitwise operations are useful for data manipulation and hardware control.

Q3. CPU Memory and I/O interaction

Aim

To observe the extn execution of a text editor, media player, and web browser simultaneously and analyse the interaction of CPU, memory, and I/O devices.

Observation

1. CPU Processed tasks
2. RAM stored data
3. I/O handled input/output

Conclusion

CPU, memory and I/O worked together to support efficient multitasking.

Figure 1: 8085 Assembly Program Before Execution in GNUSim8085

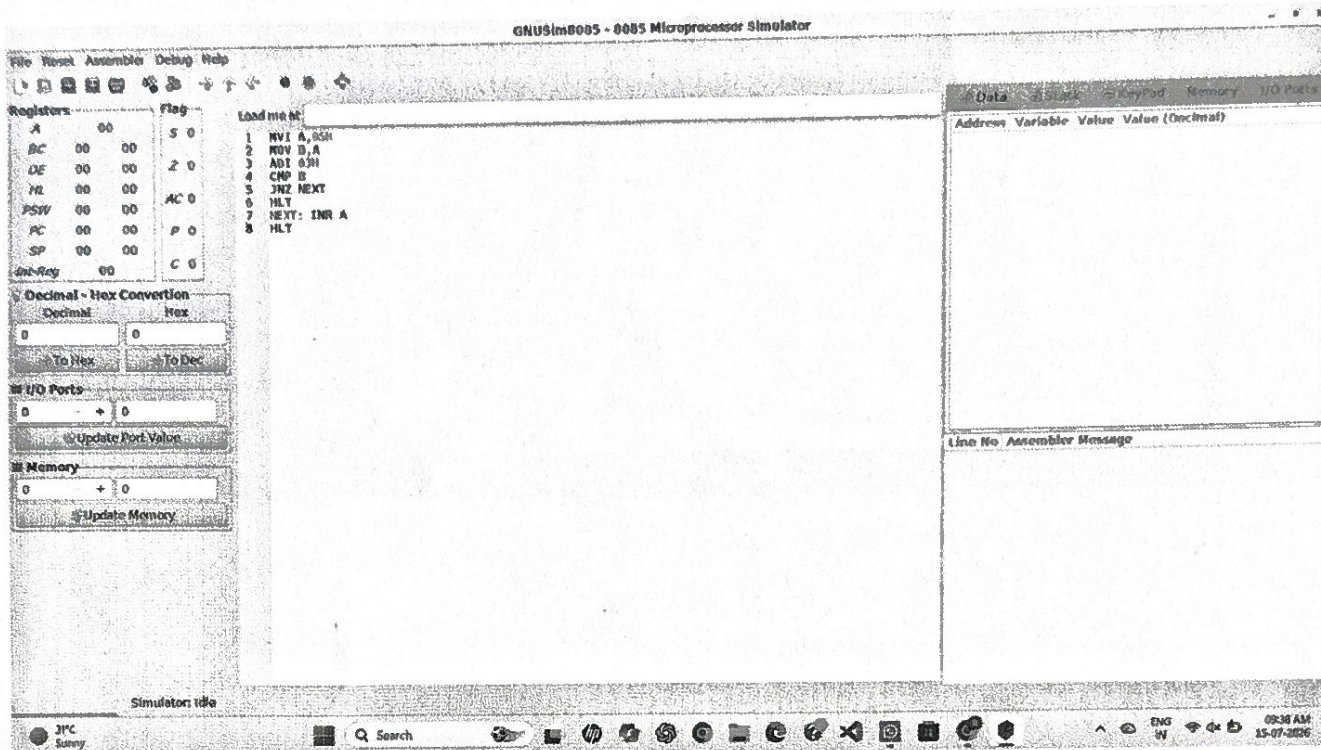


Figure 2: 8085 Program Execution Showing Register Values and Program Counter

